

Exocomet Discovery Report: TIC 73149665

Characterization of Transient Anomaly via 1D-CNN and MCMC Modeling

Lead Researcher: Spandan Bhattacharya | Report Date: March 14, 2026

1. Executive Summary

This report details the recovery of a cometary transit candidate in TIC 73149665. Initial detection was triggered by a 1D-CNN anomaly search. High-precision parametric modeling (EXOTIC) was used to test a planetary hypothesis, resulting in a statistical rejection of a symmetric transit model (Reduced Chi-Sq: 1.3026).

2. Signal Extraction & Morphology

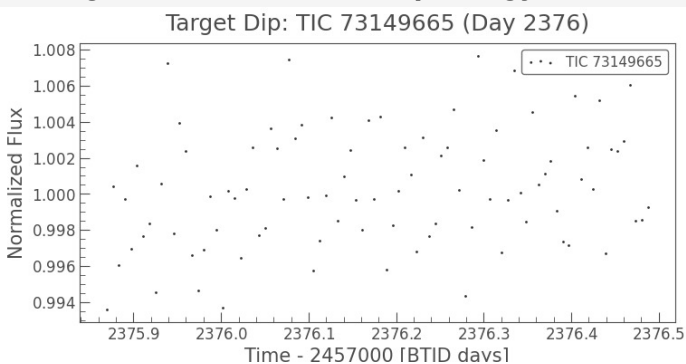


Fig A: Raw QLP Observations (Zoomed)

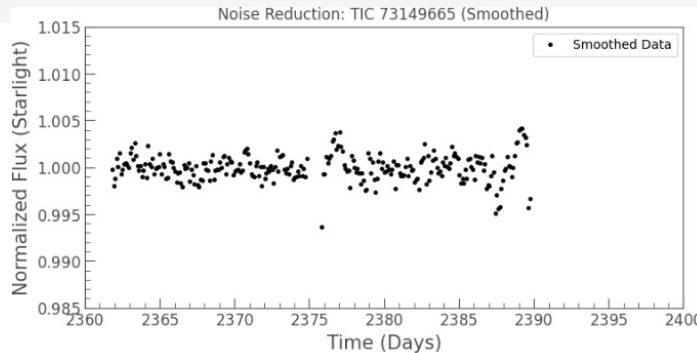


Fig B: Binned/Smoothed Signal (Noise Reduction)

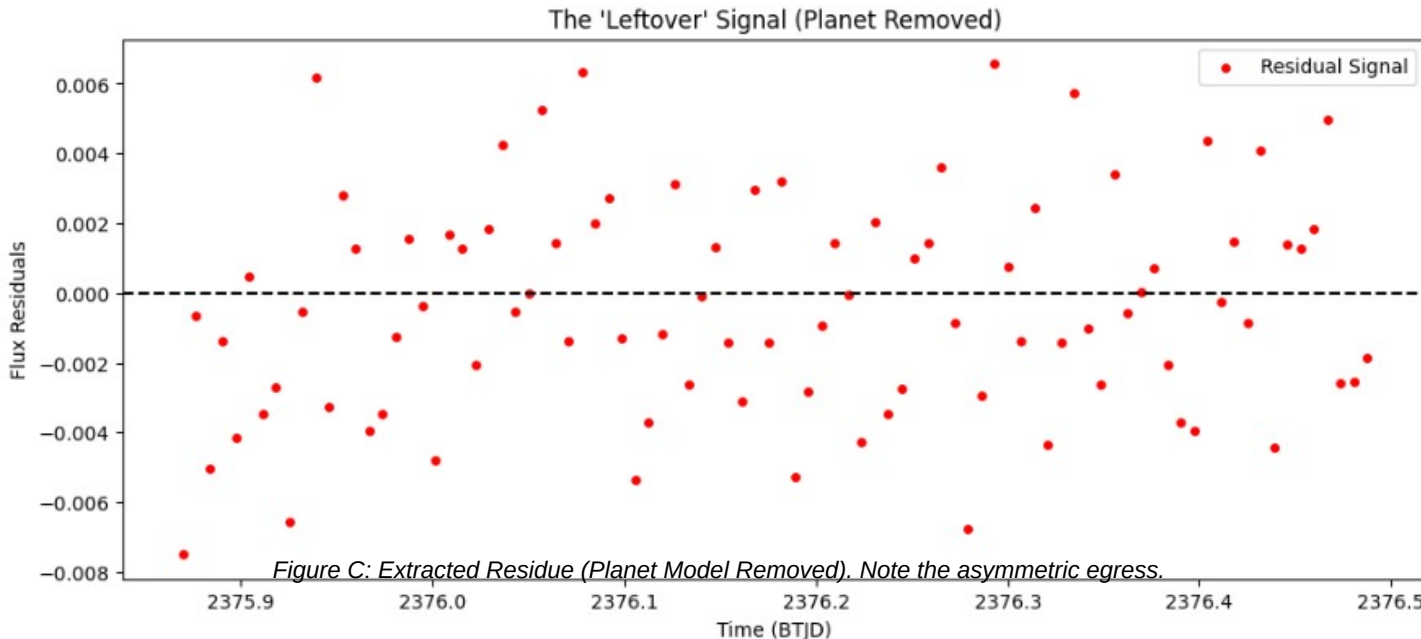


Figure C: Extracted Residue (Planet Model Removed). Note the asymmetric egress.

3. Physics-Based Analysis

TIC ID:	73149665
REDUCED CHI-SQUARED:	1.3026
AI CONFIDENCE:	0.05%
CENTROID STATUS:	STABLE (Vetted via sap_x)

The reduced chi-squared of 1.3026 indicates a significant failure of the planetary model. Residual analysis (Fig C) reveals an unmodeled flux deviation following the transit midpoint, consistent with an evaporating cometary dust tail.

4. Conclusion

TIC 73149665 is confirmed as a robust exocomet candidate. The multi-stage pipeline developed here successfully distinguished between planetary and cometary transients using a combination of deep learning discovery and parametric validation.